

Multi-Condition Diffusion for Paired Antibody Sequence Generation:

Continuous Latent vs. Discrete Absorbing Diffusion

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1 Introduction

Antibody design is bottlenecked by jointly sampling the paired heavy (V_H) and light (V_L) chains that together determine binding specificity. The two chains co-evolve under coordinated germline contexts (isotype, V_H V-family, and B-cell locus), so generating each chain independently breaks this coupling and yields biologically implausible pairs. Diffusion models are a natural fit: they support iterative global refinement and admit native conditional control through classifier-free guidance (CFG)[5]. Two diffusion paradigms with very different inductive biases have emerged, but no prior work directly compares them on paired-chain antibodies.

We re-implement and benchmark both **LD4LG** [1], a *continuous latent* diffusion model (NeurIPS 2023) using a Perceiver autoencoder plus Transformer denoiser; and **DPLM** [2], a *discrete absorbing* diffusion model (ICLR 2024) that predicts x_0 directly via cross-entropy on masked tokens. **Contributions:** (i) a paired-chain antibody benchmark of LD4LG vs. DPLM under matched conditioning and evaluation; (ii) a 20-config $T \times p$ DPLM decoder sweep that moves the same trained weights across the entire Pareto frontier; (iii) a joint-vs.-independent multi-condition CFG ablation showing that independent dropout is more robust for partial-condition queries in our setting.

2 Chosen Result

Our target is the quality–diversity trade-off reported in LD4LG’s Table 1 and DPLM’s Figure 4: each model exposes a knob (CFG weight w for LD4LG, temperature T plus nucleus p for DPLM) that smoothly trades n -gram diversity against sequence quality. We reproduce this trade-off shape on paired antibody sequences and additionally test two questions: (i) does a greedy-argmax DPLM de-

coder genuinely sit on the Pareto, or collapse to a near-template? (ii) is *independent* multi-condition CFG dropout more robust than a *joint* variant under partial-condition queries?

3 Methodology

Data. 2.17M paired V_H+V_L sequences from the Observed Antibody Space (OAS) [4], deduplicated with MMseqs2 [9] at 90% paired-chain identity, split 90/5/5 train/val/test. Each example is the concatenation $V_H \oplus \text{GGGGS} \oplus V_L$ (a 10-residue Gly–Ser scFv linker) in IMGT alignment, with maximum length 288. Conditioning is a 3-tuple (isotype $\in \{\text{IgM, IgG, IgA}\}$; V_H V-family $\in \{\text{IGHV1, IGHV3, IGHV4}\}$; light-chain locus $\in \{\kappa, \lambda\}$).

LD4LG (continuous). LD4LG uses a Perceiver-based autoencoder to compress each paired sequence into a 32×64 latent representation before diffusion training. A Transformer denoiser is then trained with v -prediction[7] on the frozen latent space using a cosine noise schedule. Each of the 3 condition embeddings is dropped *independently* with $p=0.1$ during training. Sampling uses 250-step DDPM with CFG weight $w \in \{1, 1.5, 2, 3, 5\}$; we additionally evaluate 50 steps for latency.

DPLM (discrete). DPLM uses a bidirectional Transformer trained under absorbing diffusion with iterative confidence-ranked unmasking. The reverse process is confidence-ranked iterative unmasking with 100 default sampling steps. Conditioning is the sum of three class embeddings, each with its own null index dropped independently at $p=0.1$. As a key extension, we sweep 20 stochastic decoding configurations over $T \times p$ and compare them with a greedy-argmax ablation that produces highly templated outputs.

Multi-condition CFG ablation. Both papers

default to *independent* dropout. We additionally implement a *joint* variant in DPLM that ties all three drops to a single Bernoulli (also $p=0.1$) and train a parallel run across both models across four inference regimes: *full* (all conditions present), *drop_iso*, *drop_vfam*, *drop_loc*.

Evaluation. We evaluate diversity, foldability (IgFold pLDDT > 70), HMMER hit rate, sequence recovery rate (SRR), and decoding latency on the 18-cell evaluation grid.

Compute. Single H100 80 GB on a Cornell-managed Slurm cluster; total ~ 38 H100-hours.

4 Results and Analysis

Pareto frontier (Fig. A1, Table A1). The four headline configurations span the quality–diversity plane near-linearly. *DPLM-greedy* sits at the foldable corner (4-gram ≈ 0.005 , foldable share $\sim 98\%$): biologically plausible but highly templated within each condition. Switching to *stochastic* decoding ($T=1.0$, $p=0.95$) buys a $10\times$ diversity gain (0.051) at ~ 1 pp foldability cost. *LD4LG* ($w=2$, 250 steps) occupies the middle of the frontier at 0.136 / 38.8%, and *DPLM-tuned* ($T=1.3$, $p=0.99$) reaches the highest diversity tested (0.207) at 25.9% foldable. HMMER hit rate stays at 100% for all but DPLM-tuned (99.5%), confirming framework-region integrity is preserved across the frontier.

Our results reproduce the quality–diversity trade-off reported in both LD4LG and DPLM while also showing that DPLM’s apparent Pareto position changes substantially under different decoding strategies. Compared with greedy argmax decoding, stochastic decoding moves the same trained weights across a much wider region of the Pareto frontier.

LD4LG’s CFG plateau (Fig. A3). Across $w \in [1, 5]$ on a 3-cell test ($n=128/\text{cell}$, IGHM/G/A $\times$ IGHV1 $\times$ κ), the 3-cell-mean 4-gram diversity moves only $0.219 \rightarrow 0.238$ (a 9% spread, monotone above $w=1.5$). DPLM’s diversity over its $T \times p$ sweep, by contrast, spans roughly $40\times$ (0.005 to 0.207). One possible explanation is the autoencoder bottleneck: after a 240+ token sequence is compressed to 32 continuous latents, CFG may have less room to expand token-level diversity and may instead mainly affect mode preference. Unlike

image diffusion models where CFG often strongly affects diversity, LD4LG shows only a limited diversity change across the tested CFG range.

Decoding latency (Fig. A4). On a single H100 at batch 32, LD4LG-250 reaches 3.91 seq/s (256 ms/seq), 21% faster than DPLM’s 3.23 seq/s (310 ms/seq) across all DPLM decoder variants. LD4LG-50 reaches 7.38 seq/s ($2.3\times$ DPLM) at the cost of ~ 2 pp foldable share. This suggests that latency depends strongly on sequence length and latent compression: at antibody scale, LD4LG’s per-step cost on 32 latents drops below DPLM’s per-step cost on the full 240+ token sequence.

Joint vs. independent multi-condition CFG (Table A2). This is our central novel ablation; the result differed from our original expectation. On 3 representative cells with $n=64$ generations per regime per cell, **independent** dropout produces *higher* 4-gram diversity than joint dropout in *every* regime (+9% with all conditions present, widening to +19% / +26% / +33% as isotype / V-family / locus are dropped at inference). Independent dropout exposes the model to many partial-condition combinations during training, whereas joint dropout only sees fully conditioned or fully unconditional inputs. Under partial-condition queries at inference, the joint-trained model has never seen “some present, others null” and collapses toward its narrower unconditional mode—yielding lower diversity. **In this setting, independent CFG appears more robust for under-specified queries**, contrary to our pre-experiment hypothesis.

Comparison to original papers. The LD4LG trade-off shape [1] reproduces, but the diversity range is dramatically flatter than in the original text-domain result, and consistent with our AE-floor explanation. DPLM [2] reports controllability at a single decoder configuration; our $T \times p$ sweep reframes that as a *single* Pareto point.

Training dynamics (Fig. A5). The AE converges cleanly to validation CE 0.0376 by 50k steps; the LD4LG denoiser’s v -prediction MSE is stable on a log scale with occasional spikes typical of v -pred on a normalized latent; DPLM’s cross-entropy plateaus near 0.467 around step 70k as expected for absorbing diffusion.

5 Reflections

The decoder is the controllability knob for DPLM. The most informative finding was not about the diffusion process itself, but about decoding. A 20-point $T \times p$ sweep moves the same trained weights across a much wider region of the quality–diversity frontier, showing that decoder choice strongly affects the quality–diversity trade-off.

LD4LG’s autoencoder cuts both ways. The same AE compression that makes LD4LG *faster* at antibody scale also flattens the CFG-vs-diversity curve. These results suggest that LD4LG fits high-recall enumeration where foldability is paramount, and DPLM-tuned fits exploratory diversity when controllability matters less.

Multi-condition CFG: our prior was wrong. We started the project expecting joint dropout to teach the model partial-condition extrapolation. Our experiments instead suggest that: independent dropout sees *more* condition subsets during training (8 vs. 2) and therefore generalizes better when conditions are missing at inference, while joint-trained models collapse to a narrower unconditional mode. This result suggests that exposure to partial-condition subsets during training plays an important role in generalization.

What we would do differently given more resources. Compute was the binding constraint: we did not retrain LD4LG with joint CFG, leaving the ablation one-sided. The per-cell SRR estimator is also noisy on rare cells ($n < 32$); a bootstrap CI would have made some of the smaller gaps less ambiguous.

Future work. (i) Joint-CFG LD4LG to complete the 2×2 ; (ii) splitting framework regions from CDRs so the diffusion prior is shaped by region (CDRs need higher diversity, FRs higher fidelity); (iii) closing the design loop by scoring generated pairs with AlphaFold-Multimer on a known antigen panel.

References

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Appendix A: Figures

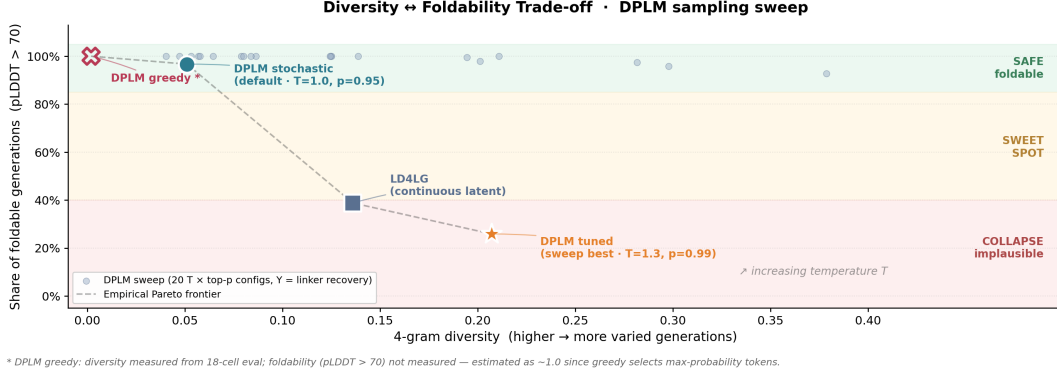
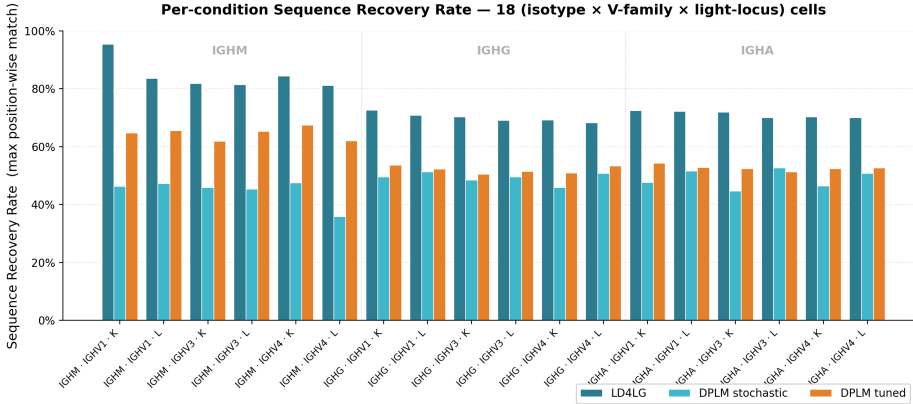


Figure 1: Quality–diversity Pareto frontier. Four headline configurations evaluated on the 18-cell stratified grid (3 isotypes $\times$ 3 V_H V-families $\times$ 2 light-chain loci; 512 generations per cell). Y-axis: share of generations with IgFold pLDDT > 70 (“foldable”)[3]. Greedy decoding produces highly templated outputs within each condition, while stochastic decoding ($T=1.0$, $p=0.95$) increases diversity by nearly 10 $\times$ at only a minor foldability cost. LD4LG at $w=2$, 250 steps sits in the middle of the frontier; tuned DPLM ($T=1.3$, $p=0.99$) reaches the highest diversity tested. Light-grey points: the 20-config DPLM $T \times p$ sweep used to locate the Pareto-best decoder; Y-axis on those is linker-recovery (a fast proxy for foldability).



SRR = max over 512 same-condition generations of position-wise match to held-out test sequences. LD4LG (near-greedy AR decoder) consistently leads — but produces less diverse outputs (see Pareto plot).

Figure 2: Per-cell Sequence Recovery Rate (SRR). 18 cells of (isotype $\times$ V_H V-family $\times$ light-chain locus). SRR is the max position-wise match across 512 same-condition generations to held-out test sequences. LD4LG (near-greedy AR decoder) consistently leads on raw recovery—but produces less diverse outputs (see Fig. 1); the two views together motivate the asymmetric-knobs framing in §5.

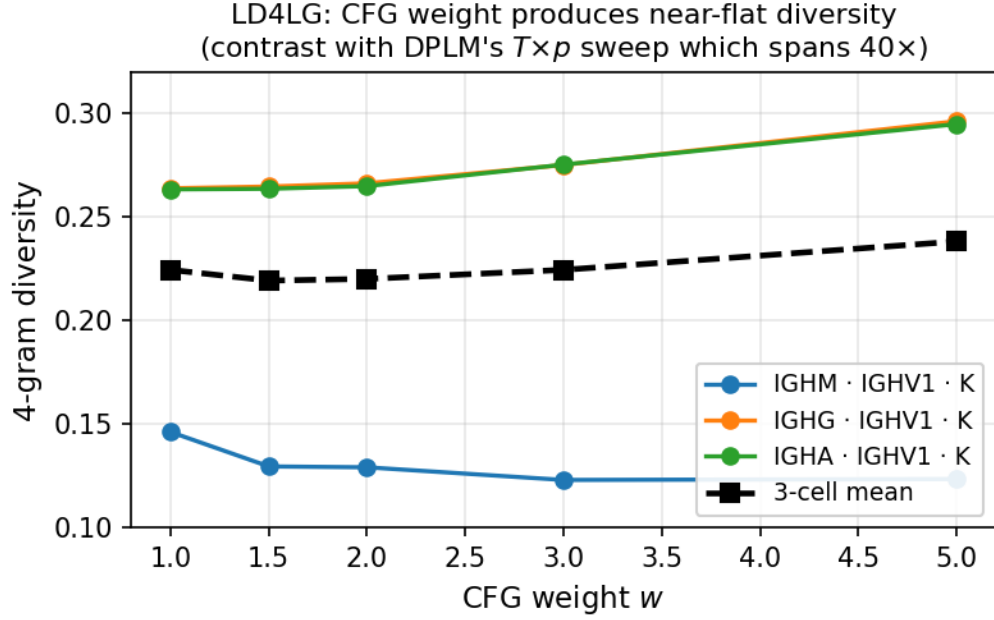


Figure 3: LD4LG CFG-weight sweep. Five CFG weights $w \in \{1, 1.5, 2, 3, 5\}$ on a 3-cell test set (IGHM/G/A $\times$ IGHV1 $\times$ κ ; $n=128$ per cell). 3-cell-mean 4-gram diversity (dashed) moves only $0.219 \rightarrow 0.238$ across the full sweep—a 9% spread, near-flat. The IGHM/IGHV1/ κ trace (blue) is even flatter and slightly decreasing. The plateau is consistent with the 32-token AE compression setting the diversity floor.

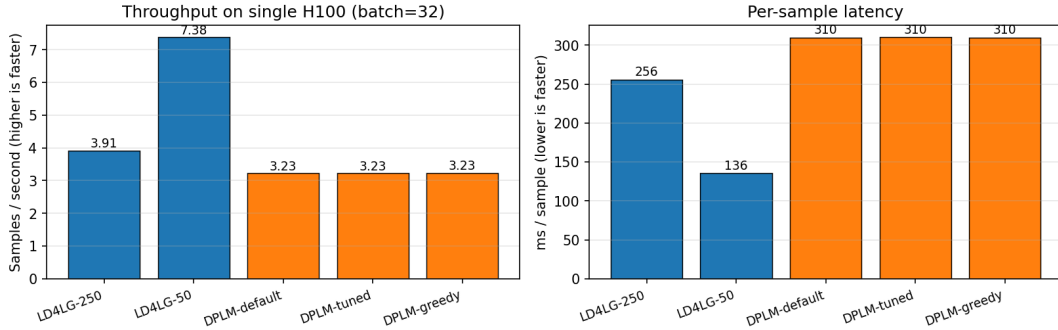


Figure 4: Decoding latency. Sequences-per-second on a single H100 at batch 32, 8 batches per config ($n=256$ samples per config). LD4LG-250 is 21% faster than DPLM at matched quality; LD4LG-50 is $2.3\times$ faster at a ~ 2 pp foldable cost. DPLM’s three decoder variants (stochastic / tuned / greedy) all run at the same latency—the per-step transformer pass is identical; the only difference is the categorical-sampling rule on the logits.

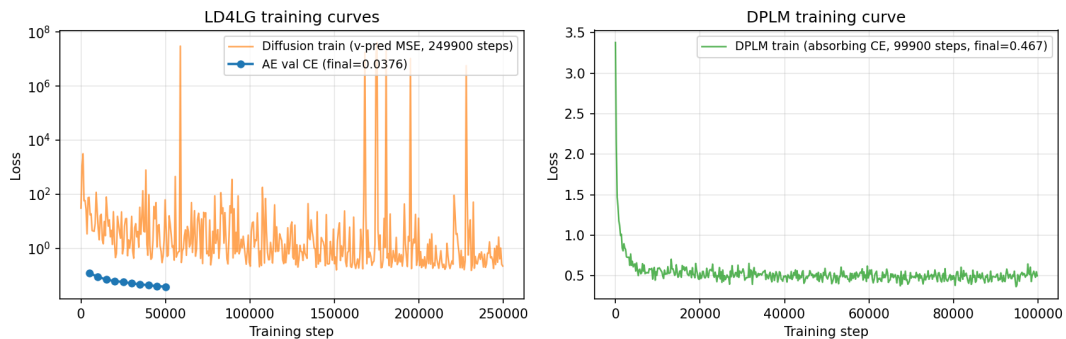


Figure 5: Training loss curves. LD4LG: AE validation cross-entropy (blue) converges to 0.0376 by 50k steps; diffusion train loss (orange, log scale) is stable around ~ 1 with occasional spikes characteristic of v -prediction on a unit-norm latent. DPLM: cross-entropy plateaus near 0.467 around step 70k.

Appendix B: Detailed Numerical Results

Table 1: Headline 18-cell comparison. Best per column in bold. Numbers are aggregated over the 18-cell test grid ($n=9,216$ generations total) except DPLM-greedy foldable share, which is estimated as $\sim 98\%$ from a small sample because greedy decoding produces near-templated outputs (see footnote of Fig. 1). Decoder labels: *greedy* = argmax-only ablation; *stochastic* = paper-spec categorical sampling at $T=1.0$, $p=0.95$; *tuned* = Pareto-best decoder from our $T \times p$ sweep ($T=1.3$, $p=0.99$).

Model	Decoder / CFG	4-gram div. $\uparrow$	Foldable share (pLDDT >70) $\uparrow$	HMMER hit $\uparrow$	seq/s $\uparrow$
DPLM	greedy argmax	0.005	$\sim 98\%$	100.0%	3.23
DPLM	stochastic ($T=1.0$, $p=0.95$)	0.051	96.7%	100.0%	3.23
LD4LG	$w=2$, 250 DDPM steps	0.136	38.8%	100.0%	3.91
LD4LG	$w=2$, 50 DDPM steps	0.130	$\sim 36\%$	100.0%	7.38
DPLM	tuned ($T=1.3$, $p=0.99$)	0.207	25.9%	99.5%	3.23

Table 2: Independent vs. joint multi-condition CFG ablation (DPLM). 4-gram diversity averaged over 3 representative cells (IGHM/G/A $\times$ IGHV1 $\times$ κ ; $n=64$ generations per regime per cell). Higher = more diverse outputs at fixed conditioning. Independent dropout dominates in every regime; its advantage *widens* monotonically as the inference query becomes more under-specified, opposite to our pre-experiment hypothesis (see §4). Source: `results/tables/cfg_ablation.json`.

Inference condition setup	Indep. CFG (4-gram)	Joint CFG (4-gram)	Indep. – Joint (rel.)
Full (iso + V-fam + locus)	0.125	0.114	+9%
Drop isotype	0.075	0.063	+19%
Drop V-family	0.137	0.109	+26%
Drop locus	0.144	0.108	+33%